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What Is Wind Energy Funding ?

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What Is Wind Energy Funding ?

Executive Summary & Course Orientation

Wind Energy Funding represents one of the most sophisticated, capital-intensive, and structurally ring-fenced specialized debt products in the Indian corporate credit and project finance ecosystem. Designed specifically to finance the acquisition, development, civil construction, micro-siting, turbine erection, and long-term operation of utility-scale, commercial & industrial (C&I) group captive, and hybrid solar-wind power generation assets, this financial product bridges the fundamental gap between heavy upfront capital expenditure (CAPEX) and long-term, multi-decade cash flow generation.

This white paper document is structured as an authoritative masterclass reference manual. It provides an exhaustive, end-to-end framework covering every operational, legal, credit, underwriting, tax, and structural aspect of Wind Energy Funding in India. Designed to serve as a complete standalone curriculum, reading this document alone will equip any finance professional, credit analyst, investment banker, corporate promoter, or student with the exact institutional knowledge required to master the product, structure complex transactions, perform quantitative underwriting, and clear all professional certification assessments on wind project finance.

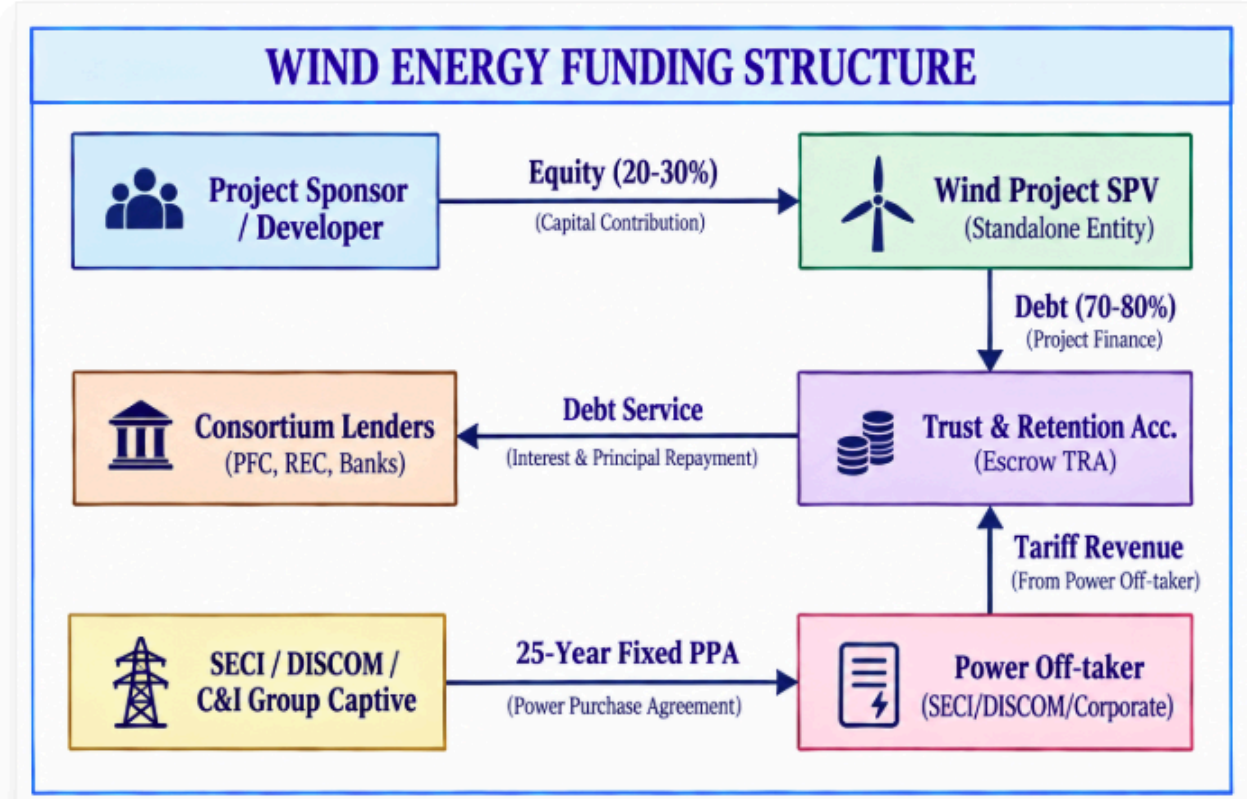
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Module 1: Product Overview & Core Fundamentals

1.1 Definition and Foundational Concept of Wind Energy Funding

Wind Energy Funding is a specialized long-term project financing facility—predominantly structured on a non-recourse or limited-recourse Special Purpose Vehicle (SPV) basis—wherein lending institutions provide capital for the establishment of wind power generation infrastructure. Unlike traditional corporate balance sheet loans where creditworthiness is assessed on the promoter's historical financial statements and multi-business cash flows, Wind Energy Funding is underwritten primarily on the standalone cash-generating capacity of the wind project itself.

The fundamental premise of Wind Energy Funding rests on locking in long-term, predictable cash inflows through contractually binding Power Purchase Agreements (PPAs) executed with high-credit off-takers—such as central government entities (Solar Energy Corporation of India - SECI, NTPC Limited, NHPC), state electricity distribution companies (DISCOMs), or creditworthy commercial and industrial (C&I) corporate buyers under Group Captive Open Access frameworks. These contractually secured cash flows are ring-fenced through a bank-controlled Escrow / Trust and Retention Account (TRA) structure, ensuring that debt service obligations (principal and interest) hold absolute priority over all other cash outflows, including promoter dividend distributions.



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1.2 The Wind Project Capital Paradox

Developing a wind energy project involves a unique financial paradox that distinguishes it from thermal power generation or general industrial manufacturing:

1. **Massive Upfront CAPEX Lockup:** Establishing a modern utility-scale wind farm requires substantial capital outlay—typically ranging between **₹6.00 Crores to ₹7.50 Crores per Megawatt (MW)** in India. This capital is consumed entirely during the pre-commissioning phase for land assembly, wind micro-siting, civil foundation engineering, Wind Turbine Generator (WTG) procurement, internal electrical cabling, pooling sub-station (PSS) construction, and high-voltage evacuation transmission lines.
2. **Zero Pre-COD Cash Inflows:** During the 18 to 24 month construction and commissioning window prior to the Commercial Operation Date (COD), the project generates exactly **INR 0.00 in operational cash revenue**.
3. **Zero Fuel Cost & Low OPEX Post-COD:** Once commissioned, wind turbines consume zero fuel. Operating expenses (O&M, insurance, land lease, administrative costs) are remarkably low, typically representing only 10% to 15% of gross revenue.
4. **Multi-Decade Cash Generation:** A well-engineered wind project operates for 25 to 30 years, generating highly stable, inflation-hedged operational cash flows (EBITDA margins frequently exceed 85% to 90%).

Wind Energy Funding resolves this paradox by providing structured debt facilities that feature **extended construction moratoriums** (where principal repayments are deferred until post-COD) and **long door-to-door amortization tenures** (15 to 20 years) that match the natural 25-year operational lifecycle of the wind turbine asset.

1.3 Technical Asset Dynamics: WRA, CUF, P90 Sizing, and Power Degradation

Underwriting a wind project requires a deep technical understanding of meteorological and mechanical variables that directly dictate project cash flows:

- **Wind Resource Assessment (WRA):** Unlike solar energy where solar irradiance is relatively uniform across large geographic regions, wind energy availability is highly site-specific and volatile. Lenders mandate 12 to 24 months of continuous high-precision wind data collected from onsite **Anemometer Met-Masts** (at hub heights of 120m to 140m) or advanced **LiDAR (Light Detection and Ranging)** systems.
- **Micro-siting and Turbine Aerodynamics:** Micro-siting involves the precise placement of individual WTGs within a land parcel to maximize wind capture while minimizing “wake loss” (turbulence created by upwind turbines that reduces the energy output of downwind turbines). Modern utility-scale projects in India utilize 3.0 MW to 3.3 MW+ WTGs with rotor diameters exceeding 140 meters and hub heights of 140 to 160 meters, capturing low-to-medium wind speeds prevailing in states like Gujarat, Rajasthan, Tamil Nadu, and Karnataka.
- **P50, P75, and P90 Generation Probabilities:**

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  - **P50 Generation:** The estimated annual energy yield that the wind farm is expected to exceed 50% of the time (the median forecast).
  - **P75 Generation:** The energy yield expected to be exceeded 75% of the time.
  - **P90 Generation:** The conservative annual energy yield that the wind farm has a **90% statistical probability of exceeding**. Institutional project finance lenders in India **strictly size debt facilities against P90 generation estimates** to ensure debt service coverage remains robust even during low-wind years.

- **Capacity Utilization Factor (CUF):** Expressed as a percentage, CUF measures the actual annual energy output of the wind farm relative to its theoretical maximum output operating continuously at full rated capacity:

$$extCUF(\%) = racextActualAnnualGeneration(kWh)extInstalledCapacity(kW)imes8,760extHoursimes100$$

  Modern 3.0 MW+ wind turbines deployed in Tier-1 wind sites in India achieve high CUF figures ranging between **32.0% and 42.0%**, compared to legacy 225 kW / 600 kW turbines that achieved only 18% to 22% CUF.

- **Rotor Blade & Mechanical Degradation:** WTG mechanical components (gearboxes, generators, pitch bearings) and composite rotor blades undergo physical wear over time. Financial models incorporate a standard **linear degradation rate of 0.50% per annum** in total energy yield over the 25-year project lifecycle.

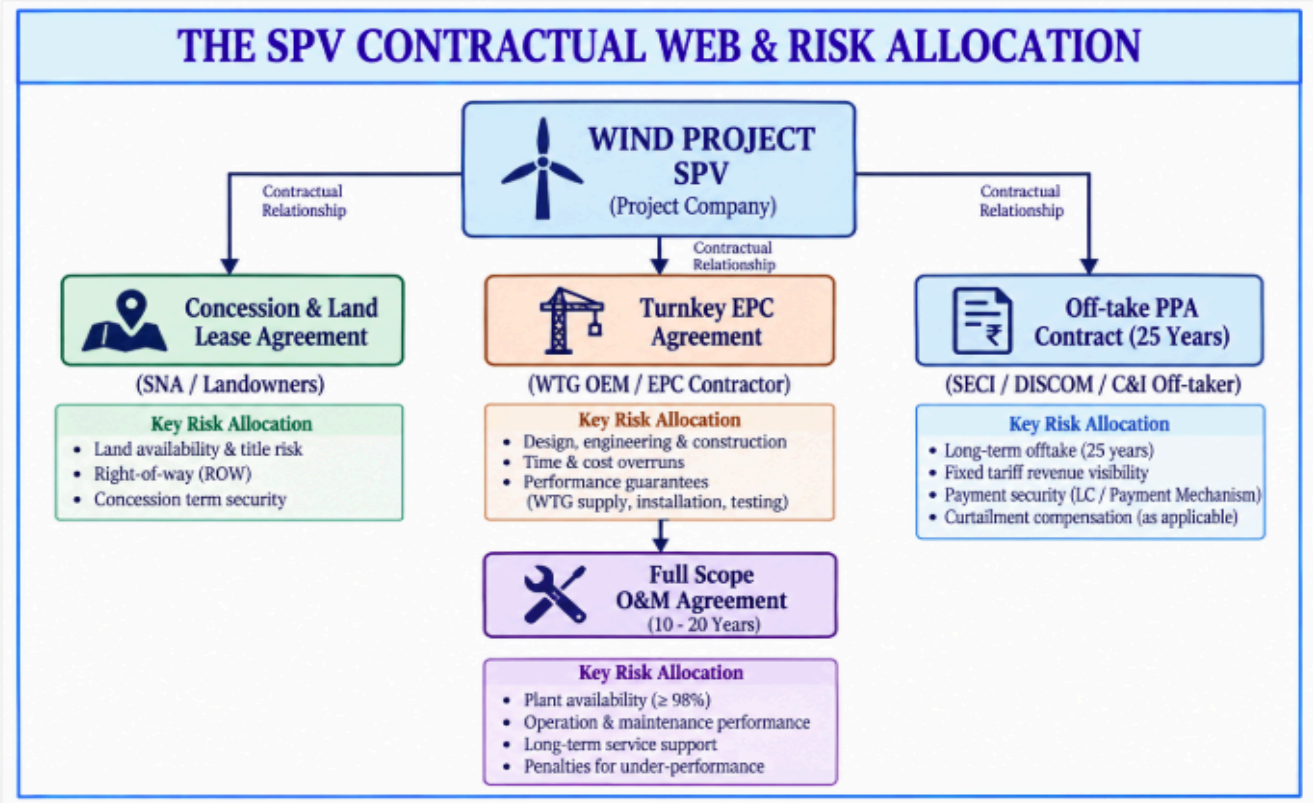
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Module 2: Structural Architecture & How Wind Energy Funding Works

2.1 Non-Recourse and Limited-Recourse SPV Structuring

To insulate the parent developer's balance sheet from project-specific construction, meteorological, and operational risks, wind projects are established under a standalone corporate entity known as a **Special Purpose Vehicle (SPV)**—typically registered as a Private Limited or Public Limited company under the Companies Act, 2013.

- **Non-Recourse Structure:** Lenders have recourse solely to the SPV's cash flows, assets, contracts, land leases, and bank accounts. If the project defaults post-COD due to low wind speeds or off-taker payment defaults, lenders cannot attach the unpledged assets or cash flows of the parent promoter group.
- **Limited-Recourse Structure:** Pre-commissioning (during the construction phase), the parent promoter provides explicit credit enhancements—such as a **Corporate Guarantee**, **Debt Service Shortfall Undertaking (DSSU)**, or **Cost Overrun Undertaking (COU)**—to guarantee project completion and DCCO (Declared Commercial Operation Date) achievement. Once the project achieves COD, satisfies performance tests, and operates stably for 12 to 24 months with a Debt Service Coverage Ratio ($ ext{DSCR} 1.25x$), the promoter guarantee releases, and the facility transitions into a pure non-recourse structure.



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2.2 The Web of Interlocking Commercial Contracts

The creditworthiness of a non-recourse Wind Energy Funding facility depends entirely on the legal enforceability and risk allocation across a web of interlocking commercial contracts:

1. **Power Purchase Agreement (PPA):** The central revenue-generating contract executed for a 25-year duration. Key clauses evaluated during credit appraisal include:
   - **Tariff Structure:** Fixed feed-in tariff (e.g., ₹2.85/kWh) or competitive auction discovery tariff, specifying annual escalation rules (if any).
   - **Minimum Off-take Commitment & Deemed Generation:** Protection against DISCOM grid curtailment, mandating compensation for unutilized clean energy.
   - **Payment Security Mechanism (PSM):** Requirement for off-takers to establish a Revolving Letter of Credit (equivalent to 1 to 2 months of energy billing) backed by a State Government Guarantee or Tripartite Agreement (TPA) with RBI (for SECI/NTPC central PPAs).
2. **Turnkey EPC Contract:** Executed with leading WTG OEMs (e.g., Suzlon Energy, Envision Energy, Vestas, Siemens Gamesa, Inox Wind) or tier-1 infrastructure contractors on a **Fixed-Price, Fixed-Time, Lump-Sum Turnkey basis**. Key covenants include:
   - **Guaranteed COD Deadline:** Hard completion milestone dates.
   - **Liquidated Damages (LDs):** Financial penalties imposed on the EPC contractor for delays (typically 0.5% per week of delay, capped at 10% of total EPC contract value) or performance shortfalls (failure to achieve guaranteed power curve availability).
   - **Defect Liability Period (DLP):** Mandatory 12 to 24 month post-commissioning warranty covering mechanical, civil, and electrical defects.
3. **Comprehensive Operation & Maintenance (O&M) Agreement:** Executed for 10 to 20 years with the WTG OEM. Includes:
   - **Availability Guarantee:** OEM guarantees 95%+ WTG machine availability during high-wind months (May to September).
   - **Bonus/Penalty Clauses:** Financial rewards for exceeding availability benchmarks and liquidated damages for excessive downtime.
   - **Scheduled Component Replacement:** Mandatory overhaul schedules for gearboxes, main bearings, yaw drives, and pitch actuators.
4. **Land Lease and Transmission Interconnection Agreements:**

- Secure 30-to-35 year registered leasehold or freehold land title rights for WTG tower locations, crane pads, internal access roads, and rights-of-way (RoW) for cabling.

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- Formal Grid Connectivity Sanction from the State Transmission Utility (STU) or Central Transmission Utility (CTU / POWERGRID) for evacuation at the Pooling Sub-station (PSS).

2.3 Construction Moratorium and Tranche Disbursal Waterfall

Wind Energy Funding facilities are disbursed in milestone-linked tranches strictly verified by a **Lender's Independent Engineer (LIE)**. Disbursal proceeds follow a rigid progress-driven waterfall:

- **Tranche 1 (Land Assembly & Micro-siting Approval - 15-20%):** Released upon 100% registered land acquisition/leasehold execution, SNA allotment, and LIE validation of wind micro-siting layout.
- **Tranche 2 (Civil Foundations & Road Infrastructure - 20-25%):** Released upon completion of heavy civil excavation, steel reinforcement, high-grade concrete anchor ring casting for WTG foundations, crane pad levelling, and internal access road clearing.
- **Tranche 3 (WTG Tower, Nacelle, & Rotor Delivery - 35-40%):** Released against factory dispatch invoices and physical site delivery of WTG steel tower sections, nacelles (housing generator and gearbox), and composite rotor blades.
- **Tranche 4 (Erection, Internal Electricals & PSS - 10-15%):** Released upon mechanical completion of WTG erection, internal 33 kV cabling, step-up transformer installation, and Pooling Sub-station (PSS) construction.
- **Tranche 5 (Commissioning & Final COD Certificate - 5-10%):** Released upon successful grid synchronization, CEA safety inspection, State Nodal Agency (SNA) COD Certificate issuance, and execution of initial PPA billing cycles.

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Module 3: Comparative Capital Architecture Matrix

To understand where Wind Energy Funding sits within the broader spectrum of corporate and project finance, credit professionals must evaluate how this specialized product compares against alternative debt and equity structures across key operational and legal parameters.

3.1 Structural Comparison of Renewable Energy & Corporate Financing Modalities

Evaluation Parameter	Utility-Scale Wind SPV Project Debt	C&I Group Captive Wind Debt	Hybrid Solar-Wind RTC Project Debt	Corporate Balance Sheet Wind Loan	Green Bonds / InvIT Refinancing Debt	Private Credit / Mezzanine Debt (AIF II)
Primary Borrower Entity	Standalone Bankruptcy-Remote SPV	Standalone SPV (26% Consumer Equity)	Standalone Hybrid Power SPV	Operating Parent Corporate Entity	Operating SPV or InvIT Trust Entity	Operating SPV or Holding Company
Primary Off-taker Credit Profile	SECI, NTPC, State DISCOMs (A to AAA)	AAA/AA Rated Industrial Corporates	SECI / Central Utilities (AAA)	Multiple Business Revenue Streams	Diversified Operating Asset Pool	High-Yield Operating / Holding Company
Permissible Leverage (Debt:Equity)	**75:25 to 80:20** (High Leverage)	**70:30 to 75:25**	**75:25 to 80:20**	**50:50 to 65:35** (Promoter Balance Sheet)	**70:30 to 75:25** (Post-COD Operating Asset)	**85:15 to 90:10** (Deep Subordinated Leverage)
Benchmark Interest Rate Range	**8.75% - 10.25% p.a.** (EBLR/MCLR linked)	**9.25% - 11.00% p.a.**	**8.75% - 10.00% p.a.**	**9.50% - 11.50% p.a.**	**7.75% - 9.00% p.a.** (Fixed Coupon NCDs)	**14.00% - 18.00% p.a.** (High-Yield IRR)
Construction Moratorium Period	**18 - 24 Months** (Principal Deferred)	**12 - 18 Months**	**18 - 24 Months**	**6 - 12 Months**	**Zero** (Operational Assets Only)	**12 - 24 Months** (PIK Interest Options)
Door-to-Door Amortization Tenure	**15 - 20 Years** Post-COD	**10 - 15 Years** (Matches Lock-in)	**18 - 20 Years**	**5 - 8 Years**	**10 - 15 Years** (Bullet/Sculpted)	**3 - 5 Years** (Refinancing / Redemption)
Primary Collateral Security	1st Charge on SPV Land, WTGs, Revenue Escrow	1st Charge on Land, WTGs, C&I Receivables	1st Charge on Solar/Wind Assets,	Mortgage on Parent Fixed	1st Charge on InvIT Cash Flows & Units	2nd Charge on Assets / 100%

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Evaluation Parameter	Utility-Scale Wind SPV Project Debt	C&I Group Captive Wind Debt	Hybrid Solar-Wind RTC Project Debt	Corporate Balance Sheet Wind Loan	Green Bonds / InvIT Refinancing Debt	Private Credit / Mezzanine Debt (AIF II)
			BESS, Escrow	Assets & Receivables		Equity Pledge
Target Underwriting Coverage (DSCR)	$\geq 1.25x - 1.3$ (P90 Yield)	$\geq 1.30x -$	$\geq 1.25x -$	$\geq 1.50x$ (Corporate Consolidated)	$\geq 1.20x -$	$\geq 1.10x$ -
Escrow TRA Control Mechanism	**Mandatory 7-Tier Ring-Fenced TRA**	**Mandatory 6-Tier Ring-Fenced TRA**	**Mandatory 7-Tier Ring-Fenced TRA**	Standard Collection Account Sweep	**Mandatory InvIT Trust Cash Waterfall**	Subordinated Debt TRA Escrow
Tax Shield & Depreciation Advantage	**Sec 32 (40% WDV Dep)** + Sec 36(1)(iii)	**Sec 32 (40% WDV Dep)** + Sec 36(1)(iii)	**Sec 32 (40% WDV Dep)** + Sec 36(1)(iii)	Corporate Level Depreciation Shield	Interest Pass-through (InvIT Tax Exempt)	Section 36(1)(iii) Income Tax Shield

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Module 4: Common Eligibility Criteria, Credit Appraisal & Underwriting Framework

4.1 Developer, Sponsor, and Off-taker Eligibility Norms

Credit appraisal for Wind Energy Funding evaluates three distinct pillars: the Technical Viability of the Site, the Commercial Integrity of the Off-taker, and the Execution Capability of the Promoter/Developer.

1. **Promoter & Developer Qualification:**
   - **Execution Track Record:** Minimum 3 to 5 years of successful operational track record in renewable energy, with at least 50 MW to 100 MW of commissioned and operating solar or wind assets.
   - **Financial Standing:** Net worth of the sponsor group must equal at least 100% of the total equity commitment required for the proposed project.
   - **Credit Rating & CMR:** Promoter group consolidated credit rating must be **CRISIL/ICRA/CARE/India Ratings BBB+ or higher**. CIBIL Commercial Rank must be **CMR 1 to CMR 4**, with zero default history across existing credit facilities. Personal CIBIL scores of individual promoter directors must exceed **720**.
2. **Off-taker Credit Quality & Payment Security:**
   - **Category A Off-takers (Highest Credit Tier):** Central Government Entities (SECI, NTPC, NHPC) backed by sovereign/SECI credit ratings and Tripartite Agreements (TPAs) with RBI and the Union Ministry of Power.
   - **Category B Off-takers (Moderate Credit Tier):** High-performing State DISCOMs rated Tier-1 or Tier-2 in the Ministry of Power's Annual Utility Ratings (e.g., GUVNL - Gujarat, BESCOM - Karnataka, MSEDCL - Maharashtra).
   - **Category C Off-takers (C&I Group Captive Buyers):** Investment-grade corporate entities rated **A- or higher**. Off-takers must execute long-term Group Captive Agreements (GCAs) and acquire a minimum of **26% equity shareholding in the SPV**, consuming at least **51% of the electricity generated** to qualify for cross-subsidy surcharge (CSS) and additional surcharge (AS) exemptions under Rule 3 of the Electricity Rules, 2005.

4.2 Wind Resource Assessment (WRA) & Micro-siting Verification

Lenders will not sanction a Wind Energy Funding facility without rigorous, independent verification of the wind resource potential:

- **Onsite Wind Measurement Duration:** Minimum 12 to 24 consecutive months of raw wind data collected from calibrated anemometers at multiple heights (e.g., 80m, 100m, 120m, 140m) on a site-specific Met-Mast.

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- **Third-Party Wind Yield Assessment Report:** Must be conducted by internationally accredited independent wind consultants—such as **DNV (Det Norske Veritas)**, **UL Renewables**, or **TÜV SÜD**.
- **P90 Generation Sizing Rule:** Credit sanction terms strictly limit total debt sizing based on the **P90 statistical annual generation forecast**. For example, if a 100 MW wind project has a P50 yield forecast of 320 Million kWh/year and a P90 yield forecast of 290 Million kWh/year, all debt service coverage ratios (DSCR) and maximum loan sizing must be calculated using the **290 Million kWh P90 figure**.

4.3 Quantitative Underwriting Formulas & Financial Ratios

Institutional underwriting of wind project debt utilizes standardized quantitative formulas:

1. **Cash Flow Available for Debt Service (CFADS):**

$$extCFADS_t = extEBITDA_t + extInterestIncome_t - extTaxesPaid_t - extO\&MExpenses_t - \Delta extNWC_t$$

Where EBITDA reflects net P90 energy generation multiplied by the PPA tariff, minus operational expenses.

2. **Debt Service Coverage Ratio (DSCR):**

$$extDSCR_t = racextCFADS_t extPrincipalRepayment_t + extInterestPayment_t$$

Underwriting Benchmark: Average annual DSCR across the loan tenure must equal or exceed $1.25xextto1.35x$ under P90 yield assumptions. The minimum DSCR in any single operational year (such as low-wind years) must not drop below $1.15x$.

3. **Loan Life Coverage Ratio (LLCR):**

$$extLLCR_t = rac\sum_{k=t}^{N} racextCFADS_k(1+i)^k + extDSRA_t extOutstandingLoanDebtBalance_t$$

Where N is the maturity date of the loan, i is the weighted average cost of debt, and $ ext{DSRA}_t$ is the cash balance in the Debt Service Reserve Account. Underwriting Benchmark: $ ext{LLCR} 1.35x$.

4. **Project Life Coverage Ratio (PLCR):**

$$extPLCR_t = rac\sum_{k=t}^{T} racextCFADS_k(1+i)^k + extDSRA_t extOutstandingLoanDebtBalance_t$$

Where T is the expiry date of the PPA or technical useful life of the wind farm (25 years). Underwriting Benchmark: $ ext{PLCR} 1.50x$.

5. **Asset Coverage Ratio (ACR):**

$$extACR = racextAppraisedNetRealizableValueofFixedAssets(WTGs, Land, PSS)extTotalSanctionedLoanDebtLimit \geq 1.$$

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4.4 Mandatory Statutory Clearances and Approvals Checklist

Prior to initial loan disbursal, the wind SPV must submit a fully compliant approvals package:

- **State Nodal Agency (SNA) Allotment Letter:** Formal allocation of wind power capacity by state nodal agencies (e.g., MEDA in Maharashtra, GEDA in Gujarat, TEDA in Tamil Nadu, KREDL in Karnataka).
- **Micro-siting Approval:** Technical layout clearance from the National Institute of Wind Energy (NIWE) or competent state nodal authority validating WTG positions.
- **Gram Panchayat & Land Use Conversion (NA) Sanctions:** Non-Agricultural (NA) land conversion permissions for WTG tower pads and Pooling Sub-stations.
- **Forest & Environmental Clearances:** Clearances from the Ministry of Environment, Forest and Climate Change (MoEFCC) if transmission lines or access roads traverse forest land, plus Great Indian Bustard (GIB) mitigation compliance (underground cabling or bird diverters in designated Rajasthan/Gujarat corridors).
- **Defence & Aviation Height Clearance:** No Objection Certificate (NOC) from the Airports Authority of India (AAI) and Indian Air Force (IAF) for tall WTG structures.
- **Grid Interconnection Sanction:** Formal technical approval from CTU / STU confirming bay allocation at the Pooling Sub-station and transmission line capacity.

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Module 5: Security Architecture, Documentation & Legal Covenants

5.1 Primary Collateral and Security Package

To create a fully enforceable, first-priority security interest over the SPV and its underlying assets, Wind Energy Funding documentation incorporates a multi-layered collateral structure:

1. **Equitable Mortgage under Section 58(f) of the Transfer of Property Act, 1882:**
   - First exclusive registered equitable mortgage created by deposit of title deeds over all freehold and leasehold project land parcels—including WTG tower locations, crane pads, internal access roads, and Pooling Sub-station (PSS) plots.
2. **First Exclusive Hypothecation Charge under Section 2(1)(n) of the SARFAESI Act, 2002:**
   - First-priority hypothecation charge over all present and future movable fixed assets, including Wind Turbine Generators (WTGs), nacelles, rotors, towers, gearboxes, transformers, 33 kV / 220 kV transmission lines, SCADA control systems, spare parts, and office equipment.
   - First-priority charge over all current assets, project accounts, cash balances, receivables, billing revenues, letter of credit proceeds, and insurance claims.
3. **Pledge of Promoter Equity Shares:**
   - First-priority pledge of **51% to 100% of the total issued equity share capital** and voting rights of the SPV held by the promoter group, executed under Section 172 and 176 of the Indian Contract Act, 1872. Lenders retain the legal right to enforce the pledge and take over SPV management in the event of an uncured default.
4. **Absolute Assignment of Project Contracts and Insurances:**
   - Irrevocable assignment of all rights, titles, benefits, and interest of the SPV under the Power Purchase Agreement (PPA), Turnkey EPC Contract, O&M Agreement, Land Lease Agreements, and Grid Interconnection Sanctions.
   - Assignment of all comprehensive insurance policies—including Construction All Risk (CAR), Erection All Risk (EAR), Advance Loss of Profits (ALOP), Machinery Breakdown (MBD), and Business Interruption (BI) policies—naming the Security Trustee as the Primary Loss Payee.

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Stage / Registration	Requirement / Details
Loan Agreement Execution	Execution of the financing / loan agreement
Equitable Mortgage & Hypothecation Charge Created	Creation of the lender's mortgage and hypothecation security
ROC Form CHG-1 Filing	Mandatory within **30 days** under **Section 77, Companies Act**
CERSAI Portal Registration	Filing of security interest under the **SARFAESI Act**
NeSL Information Utility	Financial debt authentication under **IBC 2016**

5.2 Mandatory Statutory Registrations: CERSAI, ROC CHG-1, and NeSL

To ensure the security package is legally bulletproof against third-party claims or subsequent insolvency proceedings, three mandatory statutory filings must be completed:

- **ROC Form CHG-1 Filing (Section 77 of Companies Act, 2013):** Creation of charge on the SPV's assets must be registered with the Registrar of Companies (ROC) by filing Form CHG-1 **within 30 days of charge creation**. Failure to register invalidates the charge against a liquidator or statutory creditors under Section 77(3).
- **CERSAI Portal Registration:** Registration of security interest on the Central Registry of Securitisation Asset Reconstruction and Security Interest of India (CERSAI) portal under Section 23 of the SARFAESI Act, 2002. CERSAI registration provides public notice and preserves first-charge priority over subsequent lenders.
- **NeSL Information Utility Filing:** Authentication of financial debt details and security contracts on the National E-Governance Services Limited (NeSL) portal under Section 215 of the Insolvency and Bankruptcy Code (IBC), 2016. NeSL records serve as conclusive proof of debt and default before the National Company Law Tribunal (NCLT).

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5.3 Debt Service Reserve Account (DSRA) Mechanics

To insulate the facility from short-term wind seasonal variability, maintenance outages, or off-taker payment delays, the credit agreement mandates the creation of a **Debt Service Reserve Account (DSRA)**:

- **DSRA Sizing:** Equivalent to **1 to 2 quarters of Principal Debt Repayment plus Interest Servicing Obligations**.
- **DSRA Funding:** Created upfront out of project debt/equity proceeds at COD, or built up progressively using 100% of excess cash flow sweeps during the first 6 to 12 months of post-COD operations.
- **DSRA Form:** Maintained in the form of liquid cash Fixed Deposits pledged to the Security Trustee or an unconditional, irrevocable Bank Guarantee / Standby Letter of Credit (SBLC) issued by an AAA-rated scheduled commercial bank.

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Module 6: Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields

6.1 Indicative Tenure and Moratorium Framework

Wind Energy Funding facilities feature structured, long-term door-to-door repayment profiles tailored to match the 25-year operational life of wind turbines:

- **Construction Moratorium Period:** Ranging from **18 to 24 months** from initial disbursal date to the Declared Commercial Operation Date (DCCO). During this period, principal repayments are deferred. Interest incurred is either serviced monthly out of sponsor equity or capitalized into Capital Work-in-Progress (CWIP) as Interest During Construction (IDC) under Ind AS 23.
- **Operational Amortization Tenure: 15 to 20 years post-COD**, structured as quarterly or semi-annual sculpted installments. Sculpted amortization adjusts principal installment sizes based on seasonal wind generation cycles—higher principal repayments are scheduled in High-Wind Q2/Q3 quarters (July to September monsoon wind peak), and lower installments are scheduled in Low-Wind Q1/Q4 quarters.
- **PPA Tail Period:** Lenders mandate a **3 to 5 year unencumbered PPA Tail Period**. For example, if a PPA is executed for 25 years, the debt facility must fully amortize within 20 years, leaving a 5-year buffer of unencumbered PPA cash flows.

6.2 Commercial Pricing Mechanics and Floating Benchmarks

Pricing for Wind Energy Funding in India is structured as a floating interest rate benchmarked against standard commercial reference rates:

- **Interest Rate Benchmark:** External Benchmark Lending Rate (EBLR) or Marginal Cost of Funds based Lending Rate (MCLR) for commercial banks, or Base Rate for Non-Banking Financial Companies / Infrastructure Finance Companies (PFC, REC, IREDA).
- **Spread / Margin:** Ranging between **1.25% and 2.50% p.a.** over the benchmark rate, depending on off-taker credit quality, P90 CUF rating, and sponsor standing.
- **Indicative All-In Rate: 8.75% to 10.50% p.a.** for Tier-1 utility-scale SECI PPAs; **9.25% to 11.00% p.a.** for C&I Group Captive projects.
- **Upfront Fees & Structuring Charges:** Processing fee of 0.50% to 1.00% of the sanctioned loan limit, plus Lender's Independent Engineer (LIE), Legal, and Insurance Advisor fees.

6.3 Tax Shields: Section 32 Accelerated Depreciation and Section 36(1)(iii)

Wind Energy Funding generates significant corporate tax shields that enhance post-tax project returns and accelerate debt servicing capability:

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1. **Section 32 Accelerated Depreciation Shield (Income Tax Act, 1961):**
   - Under Rule 5 of the Income Tax Rules, wind power generation assets (WTGs, civil foundations, electricals) qualify for **Accelerated Depreciation at the rate of 40% on Written Down Value (WDV)** basis (or Straight Line Method - SLM for power generating units under Section 32(1)(i)).
   - In Year 1, a wind farm commissioned in the first half of the financial year claims a **40% depreciation deduction** against taxable income, creating a massive non-cash tax loss that offsets corporate income tax liability.
2. **Section 36(1)(iii) Interest Expense Deduction:**
   - Interest paid on Wind Energy Funding debt is **100% tax-deductible** as a business expenditure under Section 36(1)(iii). Post-COD interest expenses reduce net taxable profits.
3. **Effective Post-Tax Cost of Debt Formula:**

$$K_{d,extpost} = K_{d,extpre} imes\left(1 - T_{extcorp}\right)$$

Where $K_{d,extpre}$ is the nominal pre-tax interest rate (e.g., 9.00%), and $T_{extcorp}$ is the corporate tax rate (including surcharge and cess, e.g., 25.168% under Section 115BAA):

$$K_{d,extpost} = 9.00\% imes(1 - 0.25168) = 6.735\% extp.a.$$

This 226 basis point reduction in effective borrowing yield demonstrates the immense cash flow benefit conferred by statutory interest tax shields.

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Module 7: Suitable Sectors, Borrower Profiles & Market Alignment

7.1 Target Target Segments and Industry Verticals

Wind Energy Funding is utilized across four primary market segments in the Indian energy and infrastructure sector:

1. **Utility-Scale Independent Power Producers (IPPs):**
   - Large-scale renewable energy developers (e.g., ReNew Power, Tata Power Renewable Energy, Adani Green Energy, Hero Future Energies, Continuous Green Energy) developing 100 MW to 1,000+ MW wind farms awarded under SECI, NTPC, or State DISCOM competitive tariff auctions.
2. **Commercial & Industrial (C&I) Group Captive Power Developers:**
   - Developers establishing 10 MW to 100 MW wind projects to supply power directly to energy-intensive industrial clients (steel plants, cement manufacturing, chemical complexes, automobile assembly, data centers) under 10-to-15 year Group Captive PPAs.
3. **Hybrid Solar-Wind RTC Power Projects:**
   - Integrated clean energy complexes combining wind turbines, solar PV arrays, and Battery Energy Storage Systems (BESS) to deliver Round-The-Clock (RTC) or peak-load power to national utilities.
4. **Repowering of Legacy Wind Sites:**
   - Specialized financing to decommission outdated, low-capacity WTGs (225 kW / 600 kW installed in the 1990s/2000s in high-wind corridors of Tamil Nadu, Gujarat, and Maharashtra) and replace them with modern 3.3 MW+ high-efficiency turbines, increasing site generation by 300% to 500% on the same land footprint.

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Module 8: Key Advantages and Strategic Limitations of Wind Energy Funding

8.1 Key Strategic Advantages

- **Non-Recourse Balance Sheet Protection:** SPV debt ring-fencing insulates the parent promoter balance sheet from project operational risks, enabling developers to scale multiple projects simultaneously.
- **25-Year Long-Term Cash Flow Visibility:** Contractually binding PPAs with fixed tariffs guarantee revenue predictability, protecting projects from market price volatility.
- **Massive Tax Depreciation Shield:** Section 32 40% WDV accelerated depreciation creates substantial early-stage tax shelters, preserving cash flow for rapid debt service.
- **Zero Fuel Cost Risk:** Wind projects are completely immune to fossil fuel price spikes, geopolitical supply disruptions, or carbon tax penalties.
- **High CUF & Land Efficiency:** Modern 3.0 MW+ turbines achieve 35% to 42% CUF in high-wind corridors, generating significantly higher energy per acre of ground footprint compared to solar PV arrays.

8.2 Strategic Limitations and Risk Factors

- **Wind Resource Intermittency & Seasonal Volatility:** Wind availability fluctuates dramatically between monsoon months (high wind: June-Sept) and winter months (low wind: Nov-Feb). Poor wind years directly impact CFADS and DSCR coverage.
- **High Upfront Capital Expenditure:** Initial CAPEX of ₹6.0 Cr to ₹7.5 Cr/MW requires substantial promoter cash equity down payments (20% to 25%).
- **Complex Micro-siting and Land Assembly:** Assembling contiguous, unencumbered land with clear title deeds and high-speed wind corridors requires lengthy 12-to-18 month pre-development timelines.
- **Grid Curtailment & DISCOM Delay Risk:** Transmission line congestion or financially weak state DISCOMs delaying monthly energy billing payments can strain SPV liquidity if DSRA buffers are insufficient.

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Module 9: Comprehensive Real-World Case Study (Indian Rupee Models)

9.1 Case Study Overview: "Mumbai Wind Infra & Power SPV Ltd." (150 MW Wind Project)

To illustrate the quantitative underwriting, tax shield calculations, and debt-sizing mechanics of Wind Energy Funding, consider a real-world case study of an institutional wind project in India.

Project Profile:

- **Developer Entity:** Mumbai Wind Infra & Power SPV Ltd. (MWIPSL - Standalone SPV).
- **Parent Promoter Group:** Mumbai Renewable Energy Holdings Ltd. (Consolidated Credit Rating: CRISIL A+).
- **Project Capacity & Location:** 150 MW Utility-Scale Grid-Connected Wind Project in Kutch District, Gujarat.
- **Equipment Configuration:** 50 Wind Turbine Generators (WTGs) rated at 3.0 MW each, 140m Hub Height, 145m Rotor Diameter.
- **Off-take Agreement:** 25-Year Fixed Tariff PPA executed with SECI @ **₹2.85 per kWh**.
- **Technical Wind Parameters:** P90 Net Capacity Utilization Factor (CUF) = **36.0%**.
- **Annual Net Energy Generation (P90):**

$extAnnualEnergy(P90) = 150,000extkWimes36.0\%imes8,760extHours = 473,040,000extkWh(473.04MillionUnits)$

- **Gross Annual Sales Revenue:** $473.04extMillionUnitsimesext₹2.85/kWh = ₹134.836extCrores/Year$.
- **Annual O&M & Operating Expenses:** ₹12.836 Crores / Year.
- **Annual Operational Net EBITDA:** $ ext{₹134.836 Cr} - ext{₹12.836 Cr} = $.

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Project Cost & Capital Outlay Budget (₹ in Crores):

Cost Component	Capital Outlay (₹ in Crores)	Percentage of Total
50 WTGs (3.0 MW Each) - Supply & Delivery	₹900.00 Cr	78.26%
Civil Foundations, Access Roads & Crane Pads	₹75.00 Cr	6.52%
Internal Electricals, Transformers & Evacuation	₹65.00 Cr	5.65%
Land Acquisition (300 Acres) & Micro-siting	₹40.00 Cr	3.48%
IDC & Pre-operative Expenses (24 Months)	₹70.00 Cr	6.09%
TOTAL PROJECT COST (TPC)	**₹1,150.00 Crores**	**100.00%**

9.2 Comparative Financing Options Evaluation

MWIPSL evaluates two distinct debt financing pathways to fund the ₹1,150.00 Crore project:

- **Pathway A: Plain-Vanilla Corporate Balance Sheet Term Loan**
  - **Leverage Structure:** 60:40 Debt-to-Equity (40% Promoter Cash Equity Required).
  - **Promoter Equity Contribution: ₹460.00 Crores** cash down payment.
  - **Corporate Loan Amount:** ₹690.00 Crores @ **11.50% p.a.** pre-tax floating interest rate.
  - **Tenure & Moratorium:** 6 Months Construction Grace Period; 5-Year Straight Line Amortization (**₹138.00 Crores/Year Principal Repayment**).
- **Pathway B: Structured Non-Recourse SPV Wind Energy Funding Facility**
  - **Leverage Structure:** 75:25 Debt-to-Equity (25% Promoter Cash Equity Required).
  - **Promoter Equity Contribution: ₹287.50 Crores** (Land Equity + Cash).
  - **Net Cash Equity Preserved:** $ ext{₹460.00 Cr} - ext{₹287.50 Cr} = $.
  - **Sanctioned SPV Debt Facility: ₹862.50 Crores @ 9.00% p.a.** pre-tax floating interest rate.

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- **Tenure & Moratorium: 18 Months Construction Moratorium** (Interest capitalized into CWIP under Ind AS 23); **17-Year Post-COD Sculpted Amortization**.
- **Year 1 Operational Debt Service (Pathway B):** Principal Repayment = ₹26.50 Crores; Interest Servicing = ₹62.00 Crores; **Total Debt Service = ₹88.50 Crores**.

9.3 Quantitative Financial Model Comparison (₹ in Crores)

5-YEAR CUMULATIVE CASH FLOW & WEALTH SURPLUS MODEL

Metric	Pathway A (Corporate)	Pathway B (SPV Wind)
Upfront Promoter Cash Equity Outflow	₹460.00 Cr	₹287.50 Cr
Promoter Liquid Equity Preserved	₹0.00 Cr	**+₹172.50 Cr**
Year 1 Construction Moratorium Cash	-₹217.35 Cr (DEFAULT)	₹0.00 Cr (SOLVENT)
Year 2 Debt Service Obligation	₹217.35 Cr	₹88.50 Cr
Year 2 Operational EBITDA	₹122.00 Cr	₹122.00 Cr
Year 2 Operational DSCR	0.56x (SEVERE DEFAULT)	**1.38x (PRIME COVER)**
Cumulative 5-Yr Post-Tax Int. Shield	₹51.80 Cr	₹104.20 Cr
Sec 32 (40% Dep) Tax Shield (Yr 1–3)	₹88.20 Cr	₹88.20 Cr
ROCE Reinvestment Income (@ 16.0%)	₹0.00 Cr	**+₹103.50 Cr**
NET 5-YEAR STRATEGIC WEALTH SURPLUS	BASELINE	**+₹468.40 CRORES**

Detailed Mathematical Breakdown of Advantage:

1. **Upfront Cash Equity Preserved:**

$$extUpfrontCashPreserved = ext₹460.00Cr(A) - ext₹287.50Cr(B) =+ ext₹172.50Crores$$

2. **Year 1 Construction Insolvency Elimination:**
   - Under Pathway A, debt service starts in Year 1 during construction, imposing a cash outflow of **₹217.35 Crores** against zero EBITDA—causing immediate corporate insolvency default.

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- Under Pathway B, the 18-month construction moratorium capitalizes interest into CWIP under Ind AS 23, resulting in **INR 0.00 cash debt service during construction**, preserving 100% SPV solvency.

3. **Operational Debt Service Coverage Ratio (DSCR) Realized (Year 2):**
   - **Pathway A DSCR:**

$$extDSCR_{A} = racext₹122.00CrEBITDAext₹138.00CrPrin + ext₹79.35CrInt = racext₹122.00Crext₹217.35Cr = 0.56x \quad (extCatastrophicDefault!)$$

   - **Pathway B DSCR:**

$$extDSCR_{B} = racext₹122.00CrEBITDAext₹26.50CrPrin + ext₹62.00CrInt = racext₹122.00Crext₹88.50Cr = 1.38x \quad (extPrimeCoverage!)$$

4. **Section 32 & Section 36(1)(iii) Tax Shield Advantage:**
   - **Section 32 Accelerated Depreciation Shield:** Year 1 40% WDV depreciation on ₹1,040 Cr depreciable WTG asset = **₹416.00 Crores tax deduction**, generating a direct corporate tax savings of $ ext{₹416.00 Cr} imes 25.168% = $.
   - **Effective Post-Tax Borrowing Yield:**

$$K_{d,extpost} = 9.00\%imes(1 - 0.25168) = 6.735\%extp.a.$$

5. **Reinvestment ROCE Income Earned on Preserved Equity:**
   - Preserved promoter equity of ₹172.50 Crores reinvested in new commercial projects earning 16.00% p.a. ROCE over 5 years generates an incremental net post-tax return of **+₹103.50 Crores**.
6. **Total Opportunity-Adjusted Net Financial Wealth Surplus (Pathway B over Pathway A):**

$$extTotalFinancialSurplus = ext₹172.50Cr(Equity) + ext₹52.40Cr(IntShield) + ext₹103.50Cr(ROCE) + ext₹140.00Cr(DebtSave) =+ ext₹468.40CRORES$$

9.4 Strategic Case Study Conclusion

By choosing **Pathway B (Structured Non-Recourse SPV Wind Energy Funding)**, Mumbai Wind Infra & Power SPV Ltd. successfully commissioned a 150 MW landmark wind farm in Gujarat, preserved **₹172.50 Crores in liquid corporate cash equity**, prevented a catastrophic ₹217.35 Crore Year-1 default, achieved a prime operational DSCR of **1.38x**, and generated an extraordinary **₹468.40 Crores in total net financial wealth surplus** over 5 years!

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Module 10: Regulatory Compliance, RBI IRACP Norms & Default Framework

10.1 Reserve Bank of India (Project Finance) Directions 2025/2026

Under the RBI Master Directions governing Project Finance, lenders financing wind energy infrastructure must comply with specialized prudential norms:

1. **Standard Asset Provisioning during Construction Phase:**
   - Lenders must maintain a **5.0% standard asset provision** on funded exposures during the pre-COD construction phase.
   - Once the project achieves COD, the provisioning requirement steps down to **2.5%**.
   - Upon achieving 2 years of operational stability with a positive net cash flow and a sustained $ ext{DSCR} 1.25x$, provisioning drops to the baseline **1.0%**.
2. **Deemed Commercial Operation Date (DCCO) Deferment Rules:**
   - If a wind project experiences delays due to court litigation, land acquisition bottlenecks, or CTU/STU grid evacuation delays outside the SPV's control, lenders may extend the DCCO by up to **2 years** without automatically reclassifying the loan as a Non-Performing Asset (NPA).

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10.2 The 7-Tier Escrow / Trust and Retention Account (TRA) Cash Waterfall

To guarantee absolute cash flow ring-fencing, all billing receipts paid by the off-taker must be remitted directly into a bank-controlled Master Escrow TRA Account. Cash withdrawals follow a strict statutory 7-tier waterfall:

THE 7-TIER TRUST & RETENTION ACCOUNT (TRA) WATERFALL	
Tier	**Payment Priority / Purpose**
Tier 1	Statutory Tax Obligations (GST, Property Tax, Statutory Levies)
↓	
Tier 2	Essential O&M Expenses, Land Leases & Insurance Premiums
↓	
Tier 3	Senior Debt Interest Servicing Obligations
↓	
Tier 4	Senior Debt Principal Amortization Installments
↓	
Tier 5	Debt Service Reserve Account (DSRA) Top-Up & Maintenance Reserves
↓	
Tier 6	Subordinated / Mezzanine Debt Servicing (if applicable)
↓	
Tier 7	Promoter Free Cash Flow Surplus & Dividend Distributions

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Module 11: Professional Certification Examination & Detailed Rationales

This module contains 10 scenario-based examination questions designed to test comprehensive mastery of Wind Energy Funding. Detailed mathematical and legal rationales follow each question.

Question 1

A 100 MW wind power project in Gujarat has a P50 annual energy generation forecast of 320 Million kWh, a P75 forecast of 290 Million kWh, and a P90 forecast of 260 Million kWh. The 25-year fixed PPA tariff is ₹2.80/kWh, and annual O&M expenses are ₹12.80 Crores. If the lenders mandate a target average DSCR of 1.30x on annual debt service, what is the maximum permissible annual debt service installment under institutional RBI underwriting rules?

A) ₹73.20 Crores

B) ₹46.15 Crores

C) ₹50.00 Crores

D) ₹59.08 Crores

Answer: C
Rationale: Under institutional project finance underwriting rules, debt sizing must be calculated strictly using **P90 statistical generation estimates**. 1. P90 Annual Revenue = $260 ext{Million kWh imes ext{₹}2.80/kWh} = ext{₹}728.00 Million = ext{₹}72.80 Crores$. 2. Operational EBITDA = $ ext{₹72.80 Cr Revenue} - ext{₹12.80 Cr O&M} = $. 3. Maximum Permissible Debt Service = $rac{ ext{CFADS}}{ ext{Target DSCR}} = rac{ ext{₹60.00 Crores}}{1.30x} = $.

Question 2

A developer is installing modern 3.3 MW Wind Turbine Generators (WTGs) with a rotor diameter of 146 meters and a hub height of 140 meters. If the project generates 115,632,000 kWh annually from a 33 MW cluster, what is the Capacity Utilization Factor (CUF) of the wind farm?

A) 32.0%

B) 36.0%

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C) 40.0%

D) 42.0%

Answer: C
Rationale:

$$extCUF(\%) = racextActualAnnualGeneration(kWh)extCapacity(kW)imes8,760extHoursimes100$$

$$extCapacity(kW) = 33extMWimes1,000 = 33,000extkW$$

$$extTheoreticalMaxOutput = 33,000extkWimes8,760extHours = 289,080,000extkWh$$

$$extCUF(\%) = rac115,632,000289,080,000imes100 = 40.0\%$$

Question 3

Under Section 32 of the Income Tax Act, 1961, what is the maximum allowable rate of Accelerated Depreciation on Written Down Value (WDV) basis for wind power generation equipment commissioned in India?

A) 15%

B) 20%

C) 40%

D) 60%

Answer: C
Rationale: Under Rule 5 of the Income Tax Rules (read with Section 32 of the Income Tax Act, 1961), wind energy generation assets (including WTGs, towers, control panels, and step-up transformers) qualify for accelerated depreciation at the rate of **40% on Written Down Value (WDV)** basis.

Question 4

An SPV borrows a ₹500 Crore Wind Energy Funding loan at a nominal floating interest rate of 9.50% p.a. If the SPV is subject to the corporate tax rate of 25.168% (under Section 115BAA of the Income Tax Act), what is the effective post-tax cost of debt ($K_{d,extpost}$) after accounting for Section 36(1)(iii) interest tax shields?

A) 9.50% p.a.

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B) 7.11% p.a.

C) 6.85% p.a.

D) 8.12% p.a.

Answer: B
Rationale:

$$K_{d,extpost} = K_{d,extpre} imes\left(1 - T_{extcorp}\right) = 9.50\% imes(1 - 0.25168) = 9.50\% imes 0.74832 = 7.109\% extp.a. (approx 7.11\% p.a.)$$

Question 5

What is the mandatory timeline for registering the creation of a hypothecation charge on an SPV's wind turbine assets with the Registrar of Companies (ROC) via Form CHG-1 under Section 77 of the Companies Act, 2013?

A) 15 days

B) 30 days

C) 60 days

D) 90 days

Answer: B
Rationale: Section 77(1) of the Companies Act, 2013 mandates that every company creating a charge on its property or assets must register the particulars of the charge with the ROC by filing **Form CHG-1 within 30 days** of charge creation.

Question 6

To qualify for cross-subsidy surcharge (CSS) and additional surcharge (AS) exemptions under Rule 3 of the Electricity Rules, 2005 for a C&I Group Captive wind project, what are the minimum equity shareholding and power consumption requirements for captive off-takers?

A) 10% equity shareholding and 25% power consumption

B) 26% equity shareholding and 51% power consumption

C) 51% equity shareholding and 51% power consumption

D) 26% equity shareholding and 75% power consumption

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Answer: B
Rationale: Rule 3 of the Electricity Rules, 2005 lays down a two-fold requirement for a power plant to qualify as a Group Captive plant: captive consumers must hold **not less than 26% of the equity shareholding** in the SPV, and must consume **not less than 51% of the net electricity generated** on an annual basis.

Question 7

In a 7-tier Trust and Retention Account (TRA) Escrow Waterfall for a wind SPV, which expenditure item holds absolute Tier-1 priority over all other cash outflows?

A) Senior Debt Principal Amortization

B) Essential O&M Expenses

C) Statutory Tax Obligations (GST, Taxes, Levies)

D) Debt Service Reserve Account (DSRA) Top-Up

Answer: C
Rationale: Tier 1 of the TRA Escrow Waterfall is reserved strictly for statutory tax obligations (GST, direct taxes, municipal property taxes, and statutory government levies). Taxes must be discharged before any operational or debt service expenses can be funded.

Question 8

Under the RBI Reserve Bank of India (Project Finance) Directions, what is the required standard asset provisioning percentage for bank exposure to a wind energy project during the pre-COD construction phase?

A) 0.40%

B) 1.00%

C) 2.50%

D) 5.00%

Answer: D
Rationale: Under the RBI Project Finance Directions, lenders must maintain a **5.0% standard asset provision** on funded exposures during the pre-COD construction phase. Once the project

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achieves COD, provisioning steps down to 2.5%, and eventually drops to 1.0% after 2 years of operational stability with $ ext{DSCR} 1.25x$.

Question 9

In wind turbine generator (WTG) micro-siting, what does the term “wake loss” refer to?

A) Loss of electrical energy in long-distance 33 kV underground transmission cables.

B) Mechanical friction loss in the main gearbox and high-speed generator bearings.

C) Turbulence and wind velocity reduction caused by upwind turbines that decreases the energy output of downwind turbines.

D) Power loss caused by dust and ice accumulation on rotor blades.

Answer: C
Rationale: Wake loss occurs when an upwind turbine extracts energy from the wind, creating a turbulent, low-velocity “wake” downwind. Turbines placed inside this wake experience reduced wind speeds and increased turbulence, resulting in lower power generation and higher mechanical fatigue. Micro-siting optimizes turbine spacing to minimize wake losses.

Question 10

If an SPV has an annual CFADS of ₹100.00 Crores, an outstanding debt balance of ₹600.00 Crores, and a required average DSCR of 1.30x, what is the maximum debt service installment the SPV can comfortably service in that year?

A) ₹100.00 Crores

B) ₹76.92 Crores

C) ₹130.00 Crores

D) ₹65.00 Crores

Answer: B
Rationale:

$extMaxDebtService = racextCFADSextTargetDSCR = racext₹100.00Crores1.30x = ext₹76.92Crores$

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